Supplementary material

Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis

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Table 1: PRISMA Checklist[15]

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Page 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Page 4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Page 4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Page 4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary Table 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Page 4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page 5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Page 6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Page 6
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Page 6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Page 6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Supplementary Table 5
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Page 6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Page 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Page 7-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Page 7-8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Page 7-8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Page 8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Page 10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Page 11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 11

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Table 2: MOOSE Checklist[16]

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	3
2	Hypothesis statement	-
3	Description of study outcome(s)	5
4	Type of exposure or intervention used	-
5	Type of study designs used	4
6	Study population	4-5
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	Title page
8	Search strategy, including time period included in the synthesis and key words	4
9	Effort to include all available studies, including contact with authors	4-5
10	Databases and registries searched	4
11	Search software used, name and version, including special features used (eg, explosion)	-
12	Use of hand searching (eg, reference lists of obtained articles)	5
13	List of citations located and those excluded, including justification	6, 13-17, Fig 1
14	Method of addressing articles published in languages other than English	-
15	Method of handling abstracts and unpublished studies	-
16	Description of any contact with authors	5
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	-
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	-
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	5
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	-
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	5
22	Assessment of heterogeneity	6
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	6-7
24	Provision of appropriate tables and graphics	Tables 1-4, Figs 1
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Figs 2, 3
26	Table giving descriptive information for each study included	Supplementary Table 5
27	Results of sensitivity testing (eg, subgroup analysis)	Suppl. Table 6
28	Indication of statistical uncertainty of findings	7-8
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	10
30	Justification for exclusion (eg, exclusion of non-English language citations)	10
31	Assessment of quality of included studies	-
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	9-10
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	-
34	Guidelines for future research	12
35	Disclosure of funding source	12

Table 3: Search strategy in MEDLINE

	Search terms
#1	(wuhan and (coronavirus or corona virus)).tw,kf,hw.
#2	(coronavirus* and ("19" or "2019")).tw,kf,hw.
#3	(2019 nCov or nCov 2019 or ncov19 or ncov 19 or novel coronavirus* or novel corona virus* or Severe Acute Respiratory Syndrome Coronavirus 2 or coronavirus disease 2019 or corona-virus disease 2019 or new coronavirus* or new corona-virus* or SARS-Coronavirus-2 or SARS-Coronavirus2 or SARS-Corona-Virus-2 or SARS-corona-virus2 or SARS-like coronavirus*).tw,kf,hw.
#4	(2019-novel CoV or SARS-COV-2 or SARS-COV2 or sarscov2 or sarscov-2 or coronavirus-19 or covid19 or covid-19).tw,kf,hw.
#5	((novel or new or nouveau or pandemic*) adj2 (CoV or covid or ncov or coronavirus or corona-virus)).tw,kf,hw.
#6	#1 OR #2 OR #3 OR #4 OR #5
#7	Knowledge/
#8	Attitude/
#9	Practice/
#10	Health Knowledge, Attitudes, Practice/
#11	#7 OR #8 OR #9 OR #10
#12	#6 AND #11
#13	limit 12 to english language

Table 4: Tool used for quality assessment[17]

	Criteria	**Yes**	**No**	**Others (CD, CR, NA)***
1	Was the research question or objective in this paper clearly stated?			
2	Was the study population clearly specified and defined?			
3	Was the participation rate of eligible persons at least 50%?			
4	Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study pre-specified and applied uniformly to all participants?			
5	Was a sample size justification, power description, or variance and effect estimates provided?			
6	For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?			
7	Was the time frame sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?			
8	For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome (e.g., categories of exposure or exposure measured as continuous variable)?			
9	Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?			
10	Was the exposure(s) assessed more than once over time?			
11	Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?			
12	Were the outcome assessors blinded to the exposure status of participants?			
13	Was loss to follow-up after baseline 20% or less?			
14	Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?			
	Quality rating (good, fair, or poor) (see guidance) Rater no. 1 initials: Rater no. 2 initials: Additional comments (if poor, please state why):			

∗CD: cannot determine; NA: not applicable; NR: not reported.

Table 5: Studies included in the systematic review and meta-analysis

	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
Asia	Azlan et al, May 2020[33]	Malaysia	Cross-sectional online survey	Malaysian above 18 years	4850	34.0	42.1	Good	3904	80.0	4346	90.4	3564	74.1
	Afzal et al, July 2020[25]	Pakistan	Cross-sectional online survey	Pakistani nationals aged 16 years or more	1004	NR*	37.0	Fair	665	66.2	756	75.3	654	65.2
	Hayat et al, May 2020[45]	Pakistan	Cross-sectional online survey	Pakistani residents age >15yrs	1257	16−30+	44.3	Fair	996	79.2	949	75.5	1073	85.3
	Leehang Lau et al, June 2020[56]	Philippines	Community-based Cross-sectional survey	participants of ICM's poverty alleviation program	2224	41.3	7.3	Fair	1719	82.2	1404	67.2	1328	63.5
	Rahman & Sathi, April 2020[61]	Bangladesh	Cross-sectional online survey	Bangladeshi internet users ≥18 years	441	18−30+	68.7	Poor	400	90.7	184	41.7	408	92.6
	Sari et al, June 2020[63]	Indonesia	Cross-sectional online survey	aged 18−60 years	201	35.6	46.3	Fair	192	95.5	174	86.6	NR	NR
	Yue et al, August 2020[68]	Henan, China	Cross-sectional online survey	People aged 15 years or older	517	15−60	46.2	Fair	294	56.9	482	93.3	350	67.7
	Zhong et al, March 2020[70]	China	Cross-sectional online survey	Chinese nationality aged 16 years or more	6910	33.0	34.3	Fair	6201	89.7	6492	93.9	6717	97.2
	Alabed et al, October 2020[26]	Malaysia	Cross-sectional online survey	People living in Malaysia aged 18 years or above	520	36.9	38.5	Poor	451	86.7	386	74.3	178	34.2
	Amalakanti et al, August 2020[30]	Andhra Pradesh, India	Cross-sectional online survey	South Indian aged above 15 years	1837	16−50+	56.5	Poor	1214	66.1	1286	70.0	1414	77.0
	Aqeel et al, August 2020[31]	Delhi, India	Cross-sectional online survey	Residents of Delhi	823	38.2	56.9	Poor	631	76.6	695	84.4	769	93.4
	Ashiq et al, August 2020[32]	Lahore, Pakistan	Cross-sectional online survey	People residing in Lahore	316	16−40+	46.5	Poor	260	82.3	221	69.9	181	57.2
	Ferdous et al, October 2020[40]	Bangladesh	Cross-sectional online survey	Bangladeshi residents aged 12-64 years	2017	24.4	59.8	Poor	1230	60.9	1818	90.1	1534	76.1

	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
	Gao et al, November 2020[71]	China	Cross-sectional online survey	Chinese residents 18 years and above	2136	33.1	21.9	Poor	1948	91.2	2093	98.0	2068	96.8
	Haq et al, August 2020[44]	Punjab, Pakistan	Cross-sectional online survey	People living in Pakistan >15 years	401	NR	52.3	Poor	351	87.6	NR	NR	236	58.7
	Hossain et al, September 2020[48]	Bangladesh	Cross-sectional online survey	Bangladeshi residents 13-90 years	2157	33.5	54.1	Fair	1566	72.6	1109	51.4	1716	79.5
	Hossain MJ et al, November 2020[49]	Bangladesh	Cross-sectional community-based and online study	Bangladeshi residents	1861	15−50+	64.5	Poor	1373	73.8	936	50.3	1154	62.0
	Islam et al, October 2020[50]	Bangladesh	Cross-sectional community-based study	Aged ≥18 years slum dwellers	406	44.9	53.2	Fair	146	35.9	357	87.9	332	81.7
	Narayana et al, July 2020[58]	India	Cross-sectional online survey	Indian residents aged above 15 years	2459	24.5	57.9	Good	1839	74.7	NR	NR	2167	88.1
	Rahman et al, August 2020[61]	Dhaka, Rangpur and Bogura, Bangladesh	Cross-sectional online and offline study	Adult population living in Bangladesh	1549	18−61	57.9	Poor	864	55.8	1207	77.9	1140	73.5
	Susilkumar & Vengadassalapathy, August 2020[65]	India	Cross-sectional online survey	Participants with access to a primary internet connection and who can read and write the English language	1015	20−60	49.3	Poor	822	81.0	926	91.1	892	87.8
	Wong et al, October 2020[66]	Hong Kong	Cross-sectional study	Age of 18 years or older South Asian in Hong Kong	352	38.9	40.3	Fair	190	53.8	238	67.4	234	66.5

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									Event, n	%	Event, n	%	Event, n	%
	Yang et al, November 2020[67]	China	Cross-sectional online survey	Residents who were aged 18 years and above, understood Chinese	919	18−46+	21.7	Good	783	85.2	854	92.9	776	84.4
	Nhu et al, September 2020[103]	Vietnam	Cross-sectional online survey	Vietnamese population	1999	18−59	21.7	Fair	1819	90.6	1719	86.0	1845	93.4
	Paul et al, September 2020[92]	Bangladesh	Cross-sectional online survey	Bangladeshi citizens aged 18 and older	1589	18−45+	60.5	Fair	1173	73.8	1303	81.6	1224	77.3
	Banik et al, January 2021[75]	Bangladesh	Cross-sectional online survey	Bangladeshi youth aged 18-35 years	707	25.03±4.26	57.1	Good	533	75.4	541	76.4	427	60.4
	Ladiwala et al, January 2021[81]	Pakistan	Cross-sectional online survey	aged 15 years and above who were permanent residents of Pakistan	1200	15−60+	38.3	Fair	988	82.3	631	52.5	859	71.6
	Sulistyawati et al, January 2021[99]	Indonesia	Cross-sectional online survey	Residence of Indonesia aged 18 years or older	816	18−50+	27.0	Poor	707	86.7	612	75.0	533	65.3
	Lee et al, February 2021[82]	South Korea	Cross-sectional online survey	aged 18 years or older, resident in South Korea and Korean speaker	970	47.44±14.78	48.6	Fair	680	70.2	718	74.0	844	87.0
	Xu et al, March 2021[104]	China	Cross-sectional online survey	Residents of 32 provinces of China	10195	30.2 ± 8.5	44.6	Fair	7340	72.4	8258	80.5	7952	77.5
	Qalati et al, April 2021[93]	China, India, Pakistan	Cross-sectional online survey	General public of China, India and Pakistan	1160	NR	71.7	Fair	989	85.3	731	63.1	731	63.4
Africa	Abdelhafiz et al, April 2020[23]	Egypt	Cross-sectional community-based and online survey	Egyptian adults	559	18−60+	37.7	Fair	414	74.1	424	75.8	NR	NR
	Ehoche et al, June 2020[38]	Nigeria	Cross-sectional online survey	North Central Nigeria	204	15−60	58.8	Poor	179	87.6	169	82.8	NR	NR

Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
								Event, n	%	Event, n	%	Event, n	%
Hezima et al, May 2020[46]	Khartoum, Sudan	Cross-sectional community-based and online survey	Sudanese residents	812	18−55+	54.2	Good	635	78.2	725	89.3	494	60.8
Hager et al, July 2020[43]	Egypt and Nigeria	Cross-sectional online survey	adults >17 years of all educational levels	1437	18−59+	52.5	Good	1244	86.6	863	60.0	NR	NR
Kebede et al, May 2020[52]	Jimma town, South-west Ethiopia,	Hospital-based Cross-sectional study	Visitors of medical centre	247	30.5	76.5	Fair	174	70.4	NR	NR	66	26.9
Adesegun et al, September 2020[24]	Nigeria	Cross-sectional online survey	Literate nigerian who understand english	1015	26.6	45.9	Poor	792	78.0	670	66.0	613	60.4
Kasemy et al, December 2020[51]	Egypt	Cross-sectional online survey	Egyptians	3712	23.3	47.8	Fair	2504	67.4	2856	76.9	1950	52.5
Kuhangana et al, October 2020[54]	Democratic Republic of the Congo	Cross-sectional study	adult people (>18 years) frequenting ten public markets	347	37.4	17.0	Poor	146	42.2	147	42.5	NR	NR
Ngwewondo et al, September 2020[60]	Cameroon	Cross-sectional online survey	Cameroonian residents, aged 18 years or more	1006	33.0	46.9	Fair	797	79.2	663	65.8	812	80.7
Reuben et al, July 2021[94]	North-Central Nigeria	Cross-sectional online survey	Residents of north-central Nigeria who understood the English language and were 18 years old and above	589	18−59	59.6	Fair	488	82.8	545	92.5	435	73.9
Sengeh et al, September 2020[97]	Sierra Leone	Community-based cross-sectional study	Resident aged 18 years or older	1253	18−60+	52.0	Fair	652	51.6	NR	NR	865	68.7
Okello et al, December 2020[91]	Uganda	Cross-sectional online survey	Ugandans aged ≥ 18 years	362	33.5 ± 10.4	58.6	Poor	340	93.9	186	51.3	175	48.3
Bekele et al, January 2021[78]	Ethiopia	Cross-sectional online survey	Ethiopian 18 years or above	341	18−46	80.3	Fair	269	78.9	NR	NR	177	51.5

Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
								Event, n	%	Event, n	%	Event, n	%
Desalegn et al, January 2021[79]	Addis Ababa, Ethiopia	Community-based cross-sectional study	Adult aged >18 years	839	30.30±9.25	58.0	Fair	604	72.3	604	71.6	520	62.3
Habib et al, January 2021[80]	Kano, Nigeria	Cross-sectional community-based and online survey	general population including Health Care Workers (HCW) in Kano, Nigeria	886	28.58±10.25	55.4	Fair	576	65.3	629	71.4	576	65.0
Takoudjou Dzomo et al, January 2021[101]	N'Djamena, Chad	Cross-sectional study	individuals from N'Djamena who understood French and aged 18 years or older	2269	31.04±10.96	61.5	Poor	794	34.9	1384	61.4	688	30.3
Yoseph et al, January 2021[105]	Sidama, Ethiopia	Community-based cross-sectional study	Adult population who resided in the Sidama regional state for 6 months	1214	34±10	58.5	Fair	1093	90.0	1002	82.5	789	65.0
Mohamed et al, February 2021[85]	Sudan	Cross-sectional online survey	All Sudanese, aged 18 years and more	987	30.13±9.84	55.6	Good	839	85.1	819	83.0	849	86.0
Li et al, April 2021[83]	Lilongwe, Malawi	Cross-sectional community-based survey	Malawi residents at the age of 18 years or more living in Lilongwe	580	18−55+	35.0	Fair	278	48.0	371	64.0	232	40.0
Molla and Abegaz, April 2021[86]	Woldia town, Northeast Ethiopia	Community-based cross-sectional study	All the households that live in Woldia town who were aged 18 and above	404	18−39+	50.7	Fair	317	78.5	213	52.7	156	38.6
Taddese et al, April 2021[100]	Gondar, Ethiopia	Community-based cross-sectional study	People of age 18 years and above residing in Gondar city	623	33 ± 13.24	35.5	Poor	323	51.8	NR	NR	331	53.1

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									Event, n	%	Event, n	%	Event, n	%
Middle East	Al-Hanawi et al, May 2020[12]	Saudi Arabia	Cross-sectional online survey	aged 18 years or older living in KSA	3388	18−60+	41.9	Fair	2766	81.6	3252	95.9	2942	86.8
	Alahdal et al, June 2020[27]	Riyadh, Saudi Arabia	Cross-sectional online survey	18-65 years old	1767	18−65	25.0	Good	1100	62.3	1682	95.2	1406	79.6
	Honarvar et al, June 2020[47]	Shiraz, Iran	Cross-sectional and population-based study	people aged at least 15 years	1331	36.0	47.3	Good	679	50.9	818	61.5	481	36.2
	Naser et al, April 2020[59]	Jordan, Saudi Arabia and Kuwait	Cross-sectional online survey	aged 18 years who was living either in Jordan, Saudi Arabia or Kuwait	1208	18−50+	32.8	Fair	808	66.9	NR	NR	808	67.6
	Alhazmi et al, August 2020[28]	Saudi Arabia	Cross-sectional online survey	All citizens and residents over 18 years	1513	NR	45.0	Good	1230	81.3	1310	86.6	1239	81.9
	Domiati et al, August 2020[37]	Lebanon	Cross-sectional online survey	Residents of Lebanon	410	NR	41.9	Good	308	75.0	276	67.2	NR	NR
	Elayeh et al, October 2020[39]	Jordan	Cross-sectional online survey	Adult residents of Jordan	2104	18−55+	24.6	Good	1673	79.5	1236	58.7	1385	65.8
	Ghazi et al, September 2020[42]	Iraq	Cross-sectional online survey	Adults living in Iraq	272	36·4	58.1	Good	235	86.2	220	80.7	227	83.5
	Khaled et al, September 2020[53]	Aseer region, Saudi Arabia	Cross-sectional online survey	Residents of the Aseer Region, Saudi Arabia	740	18−70	11.5	Fair	594	80.3	533	71.9	575	77.6
	Zaid et al, July-August 2020[69]	Jordan	Cross-sectional online survey	Jordanian aged 18 years	3791	18−60	26.8	Poor	2114	55.7	NR	NR	NR	NR
	Nakhostin-Ansari et al, December 2020[88]	Iran	Cross-sectional online survey	Iranian at least 15 years old	1015	35.32±11.95	42.6	Fair	897	88.3	930	91.5	952	93.8
	Shahabi et al, January 2021[98]	Hormozgan, Iran	Cross-sectional online survey	Residents of Hormozgan Province aged above 15 years	2024	33.94 ± 9.37	35.6	Good	1607	79.4	1619	80.1	1864	92.1

	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
	Tawalbeh et al, January 2021[102]	Jordan	Cross-sectional online survey	persons who aged 18 years or above, can read and write Arabic or English language	2470	18−40+	31.7	Poor	1976	80.3	2247	91.0	1929	78.1
	Abbasi-Kangevari et al, February 2021[72]	Iran	Cross-sectional online survey	Iranian >18 years old	12232	32.2±9.9	38.1	Poor	9496	77.0	11481	93.1	10630	86.2
	Al Ahdab et al, February 2021[73]	Syria	Cross-sectional online survey	Persons who were aged 16 years or more	706	16−50+	37.1	Good	417	59.0	433	61.3	521	73.8
	Al-Hussami et al, February 2021[74]	Jordan	Cross-sectional online survey	Jordanian nationality aged 18 years or older	1076	34.83±10.50	40.2	Fair	904	84.4	926	86.0	849	78.8
	Naqid et al, February 2021[89]	Kurdistan, Iraq	Community-based cross-sectional study	Kurdish population in Iraq aged >15 years	885	15−75	60.0	Poor	675	76.3	558	63.2	451	51.1
	Saeed et al, February 2021[95]	Mosul, Iraq	Cross-sectional online survey	Mosul-Iraqi individuals above the age of 18 and currently residing in Mosul	909	18−50+	38.6	Fair	782	86.1	NR	NR	754	82.9
	Omar and Amer, April 2021[90]	Egypt	Cross-sectional online survey	residents of Egypt using internet excluding the Illiterate, COVID-19 cases, and health-care provider	999	16−40+	33.3	Poor	808	80.9	794	79.5	529	52.7
	Sakr et al, April 2021[96]	Lebanon	Cross-sectional online survey	Lebanese aged 18 years or above	1882	18−71	73.7	Fair	1263	67.1	1336	71.3	1412	75.0

	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
North America	Alobuia et al, June 2020[29]	USA	Telephone survey	aged 18 year and older	2906	18−65+	48.0	Fair	2288	78.7	1502	51.7	1073	36.9
	Clements JM, May 2020[35]	USA	Cross-sectional online survey	US residents aged 18 years or older	1070	37.1	58.2	Good	838	78.3	NR	NR	536	50.1
	Czeisler et al, June 2020[36]	New York (NYC) and Los Angeles (LA), USA	Cross-sectional online survey	adults aged ≥18 years	1676	18−65+	43.9	Fair	NR	NR	USA (1286) NYC (242) LA (207)	USA (76.7) NYC (84.5) LA (80.0)	USA (920) NYC (178) LA (158)	USA (54.9) NYC (62.2) LA (61.0)
	Geldsetzer P, April 2020[41]	US and UK	Cross-sectional online survey	Adults registered with Prolific Academic Pty	US (2986) UK (2988)		US (49.1) UK (48.8)	Fair	US (2424) UK (2483)	US (81.2) UK (83.1)	NR	NR	NR	NR
	Leigh et al, October 2020[57]	Canada	Cross-sectional online survey	Adult Canadian	1996	18−65	45.2	Fair	1361	68.1	NR	NR	1143	57.2
South America	Bates et al, September 2020[34]	Ecuador	Cross-sectional online survey	Ecuadorians aged 18 or greater	2399	18−50+	37.0	Fair	1929	80.4	1331	55.5	2219	92.5
	Bates et al, November 2020[77]	Colombia	Cross-sectional online survey	People who identified as Colombians and who were aged 18 or greater	459	18−50+	28.1	Fair	353	76.8	289	63.0	424	92.3
	Bates et al, April 2021[76]	Venezuela	Cross-sectional online survey	Venezuelans and who were aged 18 or greater	3122	18−50+	24.6	Fair	2685	85.7	1592	50.6	2747	87.9
Europe	La Torre et al, June 2020[55]	Italy	Cross-sectional online survey	11-30 years	5234	11−30	36.7	Fair	3364	64.3	3707	70.8	NR	NR
	Mouchtouri et al, November 2020[87]	Greece	Telephone survey	Adult population of Greece	1858	49.2±17.4	41.2	Fair	1486	80.0	1319	71.0	1059	56.7

	Lead author, month, year	Study location	Study design	Study population	Sample size, N	Mean age/ range (years)	Male participants, N (%)	Quality of the study	Knowledge (overall good knowledge)		Attitude (overall positive attitude)		Practice (overall good practice)	
									Event, n	%	Event, n	%	Event, n	%
Oceania	Seal et al, June 2020[64]	Australia	Cross-sectional online survey	Australian adults (18 years and older)	1420	18−50+	48.0	Fair	816	57.4	NR	NR	803	56.5
Multiple continents	Masoud et al, February 2021[84]	Algeria, Brazil, Egypt, Ghana, India, Indonesia, Iraq, Ireland, Jordan, Lebanon, Libya, Morocco, Nigeria, Nepal, Palestine, Pakistan, Saudi Arabia, South Africa, Sri Lanka, Sudan, Syria, UK, USA	Cross-sectional online survey	Any citizen of the included 22 countries above the age of 18	71890	27.64±9.78	40.0	Good	49388	68.7	33429	46.5	62235	86.6

*NR-Not reported

Table 6: Subgroup analysis by country income level, continent, study period, age group, gender, education and employment status

Outcome	Study characteristics	No of studies	Estimates score (%) (95% CI*), p-value	I^2	Egger test (p-value)
Knowledge	**Income level of the country**				
	Low income countries	16	72 (63−80), <0.001	99.36%	
	Lower-middle income countries	26	73 (68−78), <0.001	99.20%	
	Upper-middle income countries	25	79 (75−83), <0.001	99.49%	
	High income countries	15	72 (68−77), <0.001	99.03%	
	Continent				
	Asia	30	76 (71−81), <0.001	99.50%	
	Africa	21	71 (64−78), <0.001	99.33%	
	Middle East	20	75 (70−80), <0.001	99.19%	
	Americas	8	79 (75−83), <0.001	97.47%	
	Europe/Oceania	3	67 (54−80), <0.001	99.38%	
	Study period				
	January-March, 2020	28	73 (69−77), <0.001	99.52%	
	April-June, 2020	46	77 (74−80), <0.001	99.25%	
	July-October, 2020	8	67 (51-82), <0.001	99.72%	
	Age group	17			
	Below 30 years		78 (74−83), <0.001	97.86%	0.044
	30 years and above		80 (75−85), <0.001	98.93%	0.022
	Gender	46			
	Male		75 (71−79), <0.001	99.10%	0.051
	Female		74 (70−78), <0.001	99.39%	0.014
	Education level	39			
	Up to 12 years		71 (66−75), <0.001	98.41%	0.674
	Above 12 years		78 (74−82), <0.001	99.58%	0.081
	Employment status	20			
	Unemployed		74 (68−79), <0.001	97.30%	0.009
	Employed		77 (72−82), <0.001	99·48%	<0.001
	Retired/students		74 (68−81), <0.001	97.59%	0.632
Attitude	**Income level of the country**				
	Low income countries	12	69 (62−76), <0.001	98.73%	
	Lower-middle income countries	24	72 (66−78), <0.001	99.45%	
	Upper-middle income countries	23	78 (72−84), <0.001	99.68%	
	High income countries	12	77 (70−84), <0.001	99.51%	
	Continent				
	Asia	28	77 (72−82), <0.001	99.58%	
	Africa	17	70 (63−77), <0.001	99.15%	
	Middle East	17	79 (72−85), <0.001	99.58%	
	Americas	7	66 (56−77), <0.001	99.35%	
	Europe/Oceania	2	71 (70−72), 1.00	0.29%	
	Study period				
	January-March, 2020	20	80 (74−86), <0.001	99.71%	
	April-June, 2020	42	73 (68−77), <0.001	99.50%	
	July-October, 2020	8	65 (57-73), <0.001	98.70%	
	Age group	14			
	Below 30 years		68 (59−78), <0.001	99.45%	<0.001
	30 years and above		73 (65−81), <0.001	99.50%	0.010
	Gender	32			
	Male		73 (68−79), <0.001	99.55%	0.087
	Female		73 (68−78), <0.001	99.69%	0.004
	Education level	27			
	Up to 12 years		71 (66−76), <0.001	98.80%	0.451
	Above 12 years		71 (65−77), <0.001	99.71%	0.027
	Employment status	15			
	Unemployed		74 (66−81), <0.001	98.34%	0.029
	Employed		75 (69−82), <0.001	99.49%	0.006
	Retired/students		74 (66−82), <0.001	98.93%	0.004
Practice	**Income level of the country**				
	Low income countries	14	55 (46−64), <0.001	99.07%	
	Lower-middle income countries	24	74 (69−79), <0.001	99.36%	
	Upper-middle income countries	22	77 (70−84), <0.001	99.81%	
	High income countries	15	66 (58−73), <0.001	99.44%	
	Continent				
	Asia	29	76 (71−81), <0.001	99.62%	
	Africa	17	57 (49−65), <0.001	99.21%	
	Middle East	18	75 (68−82), <0.001	99.64%	
	Americas	9	66 (53−79), <0.001	99.72%	

Outcome	Study characteristics	No of studies	Estimates score (%) (95% CI*), p-value	I^2	Egger test (p-value)
	Europe/Oceania	2	57 (55−59), 1.00	0.28%	
	Study period				
	January-March, 2020	23	67 (60−75), <0.001	99.79%	
	April-June, 2020	42	74 (70−79), <0.001	99.58%	
	July-October, 2020	8	57 (43−72), <0.001	99.65%	
	Age group	14			
	Below 30 years		80 (72−88), <0.001	99.34%	0.025
	30 years and above		82 (74−90), <0.001	99.70%	0.006
	Gender	39			
	Male		75 (69−80), <0.001	99.63%	<0.001
	Female		77 (72−82), <0.001	99.79%	<0.001
	Education level	31			
	Up to 12 years		74 (68−79), <0.001	99.24%	0.005
	Above 12 years		77 (71−83), <0.001	99.86%	<0.001
	Employment status	18			
	Unemployed		78 (72−85), <0.001	98.79%	<0.001
	Employed		76 (68−83), <0.001	99.70%	<0.001
	Retired/students		76 (68−84), <0.001	98.95%	0.174

*CI: Confidence Interval

Table 7: Analysis by some important questions regarding knowledge, attitude and practice on COVID-19

Outcome	Study characteristics	No of studies	Estimates score (%) (95% CI*), p-value	I^2	Egger test (p-value)
Knowledge	Q1.Main symptoms	61	85 (81−89), <0.001	99.76%	0.0001
	Q2.Population at risk	46	81 (77−86), <0.001	99.77%	<0.001
	Q3.Route of transmission	58	83 (79−87), <0.001	99.78%	<0.001
	Q4.Spread of infection by asymptomatic patients	39	75 (69−81), <0.001	99.80%	<0.001
	Q5.Face mask as a preventive measure	35	70 (63−78), <0.001	99.85%	0.699
	Q6.Avoidance of crowded places as a preventive measure	27	95 (93−97), <0.001	99.20%	0.030
	Q7.Isolation and treatment of patients to prevent infection	32	91 (87−95), <0.001	99.67%	0.012
	Q8.Quarantine/Incubation period	44	87 (82−91), <0.001	99.80%	0.041
Attitude	Q1.COVID-19 will be successfully controlled	22	71 (64−78), <0.001	99.78%	0.0003
	Q2.It is important to maintain social distance	11	88 (81−94), <0.001	99.25%	0.623
Practice	Q1.Hand hygiene	49	80 (75−86), <0.001	99.82%	0.016
	Q2.Wearing face mask	56	65 (58−73), <0.001	99.90%	0.003
	Q3.Social distancing	32	70 (63−77), <0.001	99.79%	0.0001
	Q4.Avoiding crowded places	45	75 (70−81), <0.001	99.77%	0.0001

*CI: Confidence Interval

Figure 1: Correlation between components of KAP

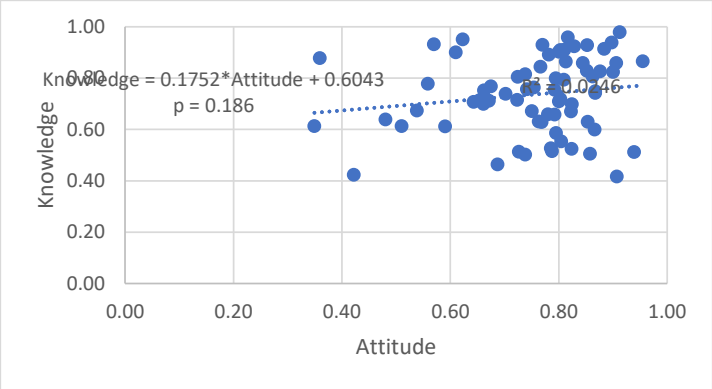


Figure 1a: Correlation between knowledge and attitude

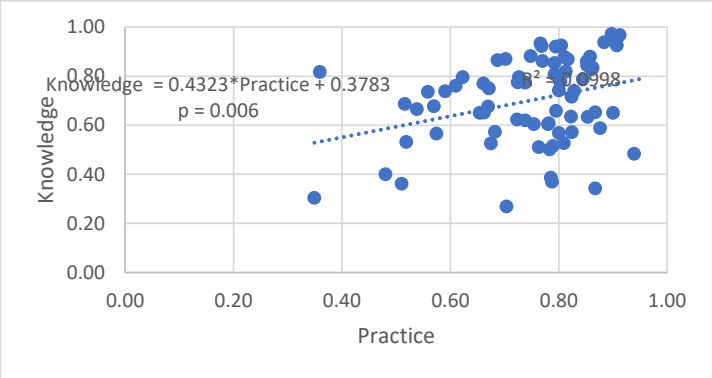


Figure 1b: Correlation between knowledge and practice

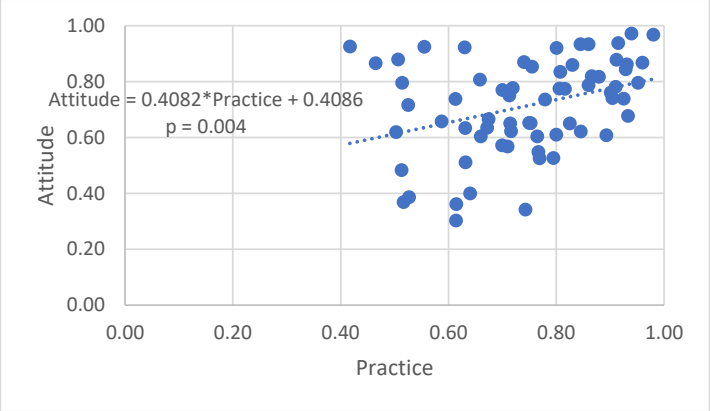
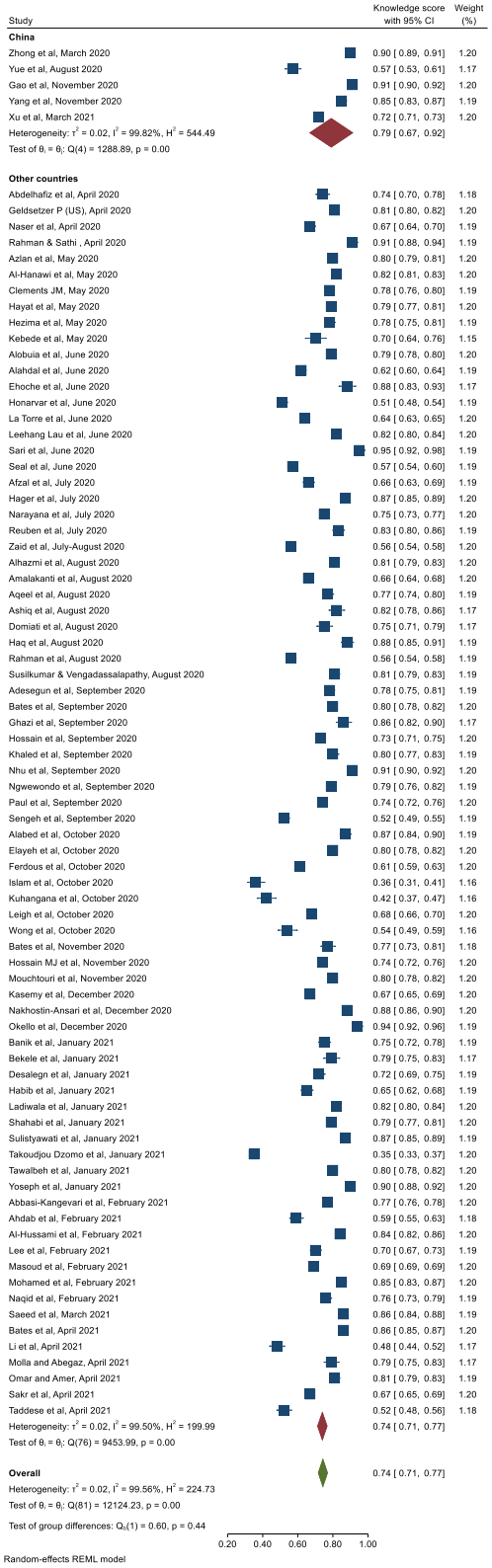
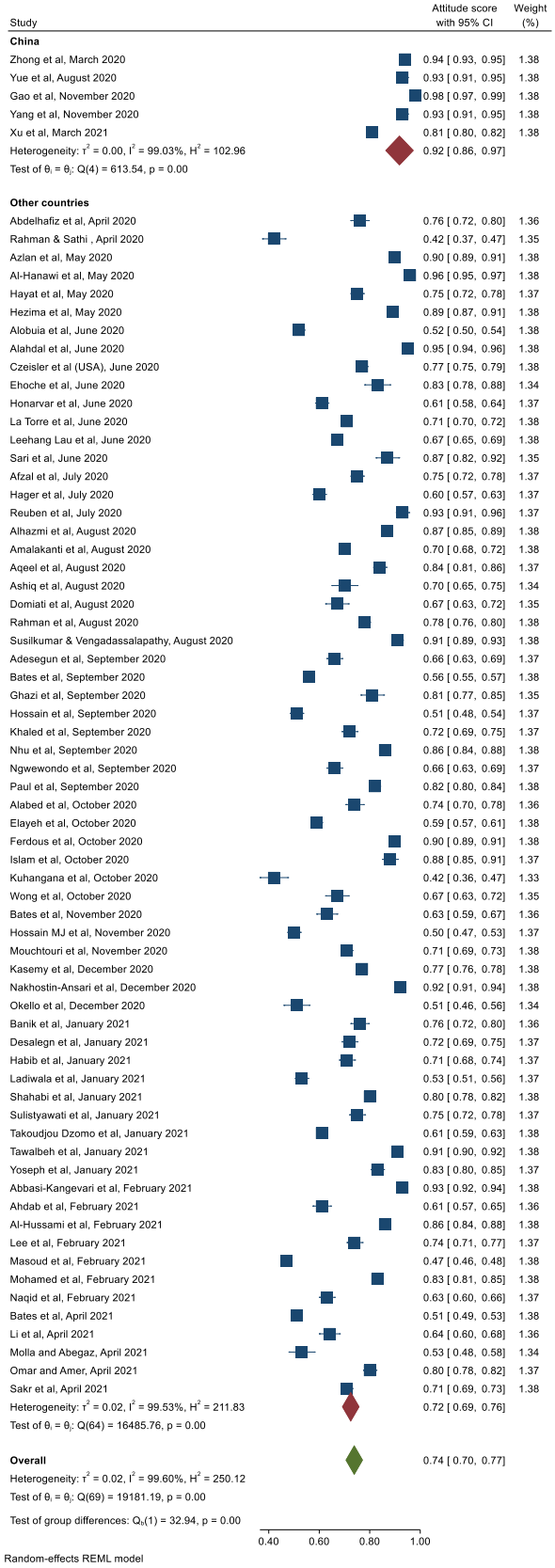


Figure 1c: Correlation between attitude and practice

Figure 2: China vs Other countries




Study
Attitude score with 95% CI
Weight (%)
China
Zhong et al, March 2020 0.94 [0.93, 0.95] 1.38
Yue et al, August 2020 0.93 [0.91, 0.95] 1.38
Gao et al, November 2020 0.98 [0.97, 0.99] 1.38
Yang et al, November 2020 0.93 [0.91, 0.95] 1.38
Xu et al, March 2021 0.81 [0.80, 0.82] 1.38
Heterogeneity: τ^2 = 0.00, I^2 = 99.03%, H^2 = 102.96 0.92 [0.86, 0.97]
Test of $\theta_i = \theta_j$: Q(4) = 613.54, p = 0.00
Other countries
Abdelhafiz et al, April 2020 0.76 [0.72, 0.80] 1.36
Rahman & Sathi , April 2020 0.42 [0.37, 0.47] 1.35
Azlan et al, May 2020 0.90 [0.89, 0.91] 1.38
Al-Hanawi et al, May 2020 0.96 [0.95, 0.97] 1.38
Hayat et al, May 2020 0.75 [0.72, 0.78] 1.37
Hezima et al, May 2020 0.89 [0.87, 0.91] 1.38
Alobuia et al, June 2020 0.52 [0.50, 0.54] 1.38
Alahdal et al, June 2020 0.95 [0.94, 0.96] 1.38
Czeisler et al (USA), June 2020 0.77 [0.75, 0.79] 1.38
Ehoche et al, June 2020 0.83 [0.78, 0.88] 1.34
Honarvar et al, June 2020 0.61 [0.58, 0.64] 1.37
La Torre et al, June 2020 0.71 [0.70, 0.72] 1.38
Leehang Lau et al, June 2020 0.67 [0.65, 0.69] 1.38
Sari et al, June 2020 0.87 [0.82, 0.92] 1.35
Afzal et al, July 2020 0.75 [0.72, 0.78] 1.37
Hager et al, July 2020 0.60 [0.57, 0.63] 1.37
Reuben et al, July 2020 0.93 [0.91, 0.96] 1.37
Alhazmi et al, August 2020 0.87 [0.85, 0.89] 1.38
Amalakanti et al, August 2020 0.70 [0.68, 0.72] 1.38
Aqeel et al, August 2020 0.84 [0.81, 0.86] 1.37
Ashiq et al, August 2020 0.70 [0.65, 0.75] 1.34
Domiati et al, August 2020 0.67 [0.63, 0.72] 1.35
Rahman et al, August 2020 0.78 [0.76, 0.80] 1.38
Susilkumar & Vengadassalapathy, August 2020 0.91 [0.89, 0.93] 1.38
Adesegun et al, September 2020 0.66 [0.63, 0.69] 1.37
Bates et al, September 2020 0.56 [0.55, 0.57] 1.38
Ghazi et al, September 2020 0.81 [0.77, 0.85] 1.35
Hossain et al, September 2020 0.51 [0.48, 0.54] 1.37
Khaled et al, September 2020 0.72 [0.69, 0.75] 1.37
Nhu et al, September 2020 0.86 [0.84, 0.88] 1.38
Ngwewondo et al, September 2020 0.66 [0.63, 0.69] 1.37
Paul et al, September 2020 0.82 [0.80, 0.84] 1.38
Alabed et al, October 2020 0.74 [0.70, 0.78] 1.36
Elayeh et al, October 2020 0.59 [0.57, 0.61] 1.38
Ferdous et al, October 2020 0.90 [0.89, 0.91] 1.38
Islam et al, October 2020 0.88 [0.85, 0.91] 1.37
Kuhangana et al, October 2020 0.42 [0.36, 0.47] 1.33
Wong et al, October 2020 0.67 [0.63, 0.72] 1.35
Bates et al, November 2020 0.63 [0.59, 0.67] 1.36
Hossain MJ et al, November 2020 0.50 [0.47, 0.53] 1.37
Mouchtouri et al, November 2020 0.71 [0.69, 0.73] 1.38
Kasemy et al, December 2020 0.77 [0.76, 0.78] 1.38
Nakhostin-Ansari et al, December 2020 0.92 [0.91, 0.94] 1.38
Okello et al, December 2020 0.51 [0.46, 0.56] 1.34
Banik et al, January 2021 0.76 [0.72, 0.80] 1.36
Desalegn et al, January 2021 0.72 [0.69, 0.75] 1.37
Habib et al, January 2021 0.71 [0.68, 0.74] 1.37
Ladiwala et al, January 2021 0.53 [0.51, 0.56] 1.37
Shahabi et al, January 2021 0.80 [0.78, 0.82] 1.38
Sulistyawati et al, January 2021 0.75 [0.72, 0.78] 1.37
Takoudjou Dzomo et al, January 2021 0.61 [0.59, 0.63] 1.38
Tawalbeh et al, January 2021 0.91 [0.90, 0.92] 1.38
Yoseph et al, January 2021 0.83 [0.80, 0.85] 1.37
Abbasi-Kangevari et al, February 2021 0.93 [0.92, 0.94] 1.38
Ahdab et al, February 2021 0.61 [0.57, 0.65] 1.36
Al-Hussami et al, February 2021 0.86 [0.84, 0.88] 1.38
Lee et al, February 2021 0.74 [0.71, 0.77] 1.37
Masoud et al, February 2021 0.47 [0.46, 0.48] 1.38
Mohamed et al, February 2021 0.83 [0.81, 0.85] 1.38
Naqid et al, February 2021 0.63 [0.60, 0.66] 1.37
Bates et al, April 2021 0.51 [0.49, 0.53] 1.38
Li et al, April 2021 0.64 [0.60, 0.68] 1.36
Molla and Abegaz, April 2021 0.53 [0.48, 0.58] 1.34
Omar and Amer, April 2021 0.80 [0.78, 0.82] 1.37
Sakr et al, April 2021 0.71 [0.69, 0.73] 1.38
Heterogeneity: τ^2 = 0.02, I^2 = 99.53%, H^2 = 211.83 0.72 [0.69, 0.76]
Test of $\theta_i = \theta_j$: Q(64) = 16485.76, p = 0.00
Overall 0.74 [0.70, 0.77]
Heterogeneity: τ^2 = 0.02, I^2 = 99.60%, H^2 = 250.12
Test of $\theta_i = \theta_j$: Q(69) = 19181.19, p = 0.00
Test of group differences: $Q_b(1)$ = 32.94, p = 0.00
0.40
0.60
0.80
1.00
Random-effects REML model

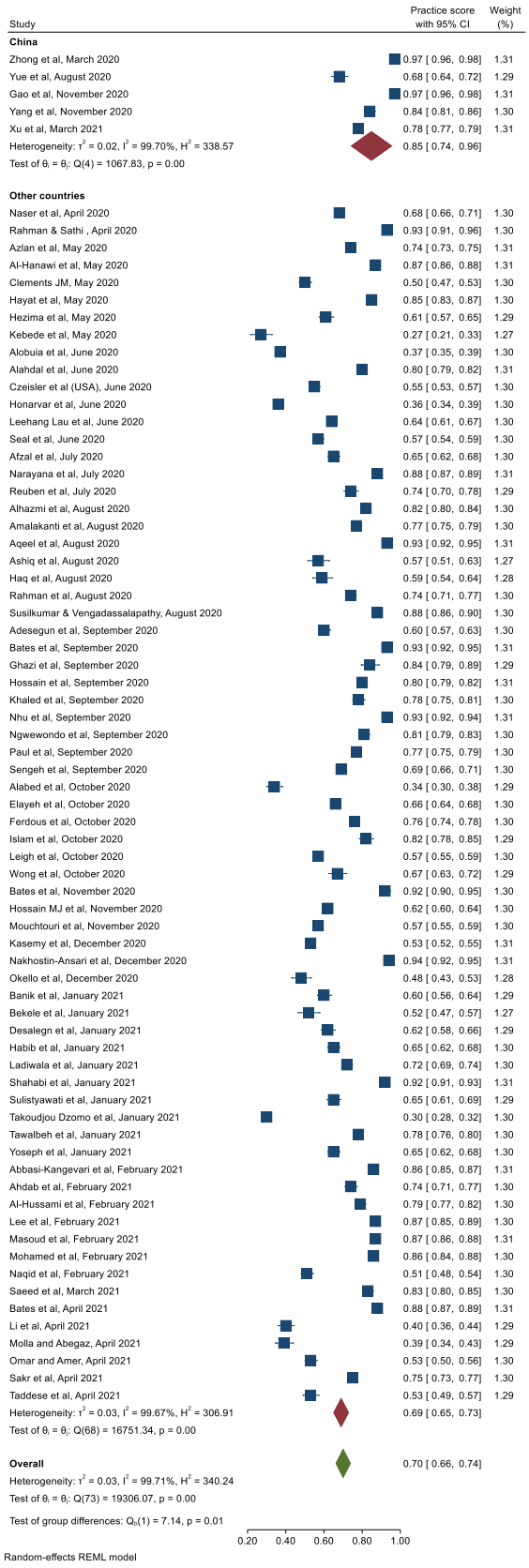
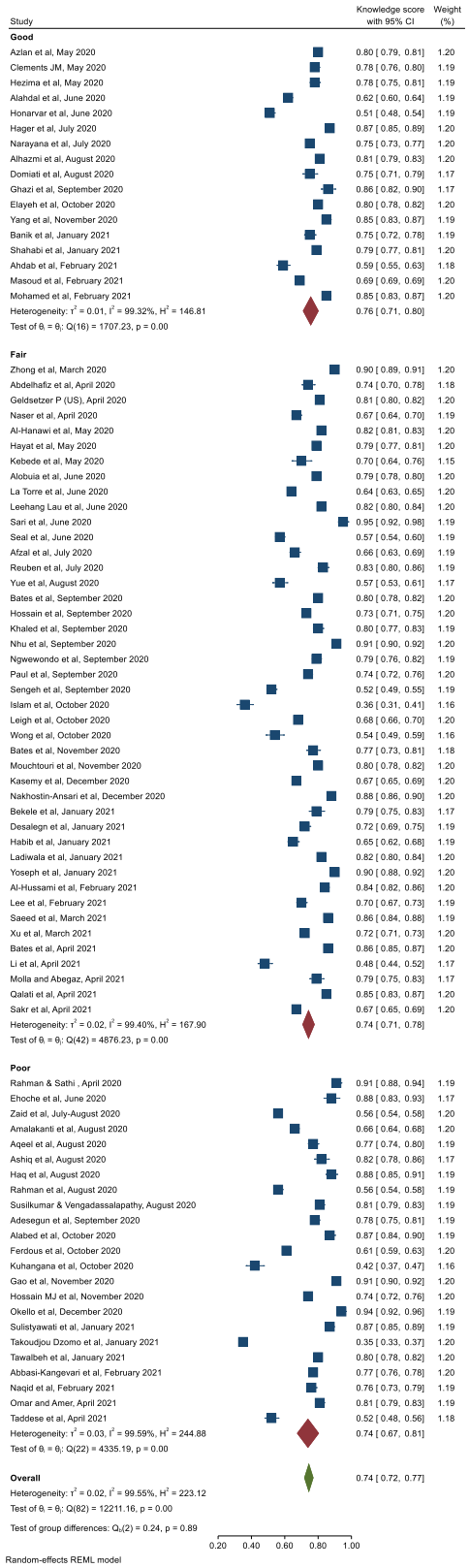
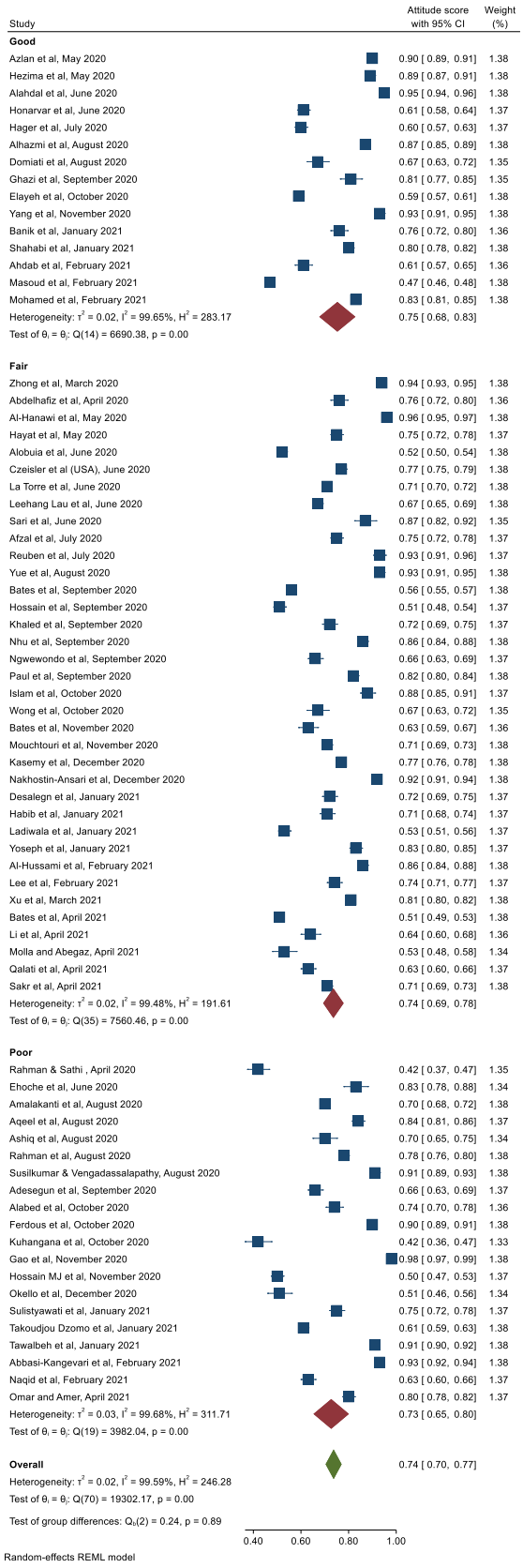
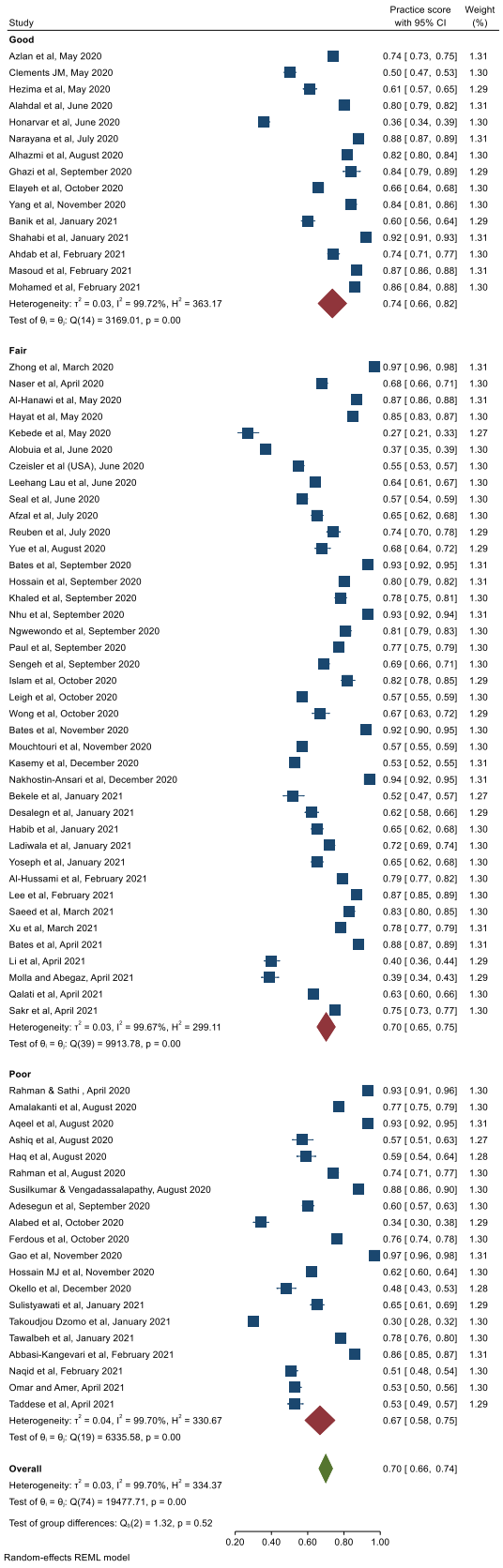

Study
Practice score with 95% CI
Weight (%)
China
Zhong et al, March 2020 0.97 [0.96, 0.98] 1.31
Yue et al, August 2020 0.68 [0.64, 0.72] 1.29
Gao et al, November 2020 0.97 [0.96, 0.98] 1.31
Yang et al, November 2020 0.84 [0.81, 0.86] 1.30
Xu et al, March 2021 0.78 [0.77, 0.79] 1.31
Heterogeneity: τ² = 0.02, I² = 99.70%, H² = 338.57 0.85 [0.74, 0.96]
Test of θi = θj: Q(4) = 1067.83, p = 0.00
Other countries
Naser et al, April 2020 0.68 [0.66, 0.71] 1.30
Rahman & Sathi , April 2020 0.93 [0.91, 0.96] 1.30
Azlan et al, May 2020 0.74 [0.73, 0.75] 1.31
Al-Hanawi et al, May 2020 0.87 [0.86, 0.88] 1.31
Clements JM, May 2020 0.50 [0.47, 0.53] 1.30
Hayat et al, May 2020 0.85 [0.83, 0.87] 1.30
Hezima et al, May 2020 0.61 [0.57, 0.65] 1.29
Kebede et al, May 2020 0.27 [0.21, 0.33] 1.27
Alobuia et al, June 2020 0.37 [0.35, 0.39] 1.30
Alahdal et al, June 2020 0.80 [0.79, 0.82] 1.31
Czeisler et al (USA), June 2020 0.55 [0.53, 0.57] 1.30
Honarvar et al, June 2020 0.36 [0.34, 0.39] 1.30
Leehang Lau et al, June 2020 0.64 [0.61, 0.67] 1.30
Seal et al, June 2020 0.57 [0.54, 0.59] 1.30
Afzal et al, July 2020 0.65 [0.62, 0.68] 1.30
Narayana et al, July 2020 0.88 [0.87, 0.89] 1.31
Reuben et al, July 2020 0.74 [0.70, 0.78] 1.29
Alhazmi et al, August 2020 0.82 [0.80, 0.84] 1.30
Amalakanti et al, August 2020 0.77 [0.75, 0.79] 1.30
Aqeel et al, August 2020 0.93 [0.92, 0.95] 1.31
Ashiq et al, August 2020 0.57 [0.51, 0.63] 1.27
Haq et al, August 2020 0.59 [0.54, 0.64] 1.28
Rahman et al, August 2020 0.74 [0.71, 0.77] 1.30
Susilkumar & Vengadassalapathy, August 2020 0.88 [0.86, 0.90] 1.30
Adesegun et al, September 2020 0.60 [0.57, 0.63] 1.30
Bates et al, September 2020 0.93 [0.92, 0.95] 1.31
Ghazi et al, September 2020 0.84 [0.79, 0.89] 1.29
Hossain et al, September 2020 0.80 [0.79, 0.82] 1.31
Khaled et al, September 2020 0.78 [0.75, 0.81] 1.30
Nhu et al, September 2020 0.93 [0.92, 0.94] 1.31
Ngwewondo et al, September 2020 0.81 [0.79, 0.83] 1.30
Paul et al, September 2020 0.77 [0.75, 0.79] 1.30
Sengeh et al, September 2020 0.69 [0.66, 0.71] 1.30
Alabed et al, October 2020 0.34 [0.30, 0.38] 1.29
Elayeh et al, October 2020 0.66 [0.64, 0.68] 1.30
Ferdous et al, October 2020 0.76 [0.74, 0.78] 1.30
Islam et al, October 2020 0.82 [0.78, 0.85] 1.29
Leigh et al, October 2020 0.57 [0.55, 0.59] 1.30
Wong et al, October 2020 0.67 [0.63, 0.72] 1.29
Bates et al, November 2020 0.92 [0.90, 0.95] 1.30
Hossain MJ et al, November 2020 0.62 [0.60, 0.64] 1.30
Mouchtouri et al, November 2020 0.57 [0.55, 0.59] 1.30
Kasemy et al, December 2020 0.53 [0.52, 0.55] 1.31
Nakhostin-Ansari et al, December 2020 0.94 [0.92, 0.95] 1.31
Okello et al, December 2020 0.48 [0.43, 0.53] 1.28
Banik et al, January 2021 0.60 [0.56, 0.64] 1.29
Bekele et al, January 2021 0.52 [0.47, 0.57] 1.27
Desalegn et al, January 2021 0.62 [0.58, 0.66] 1.29
Habib et al, January 2021 0.65 [0.62, 0.68] 1.30
Ladiwala et al, January 2021 0.72 [0.69, 0.74] 1.30
Shahabi et al, January 2021 0.92 [0.91, 0.93] 1.31
Sulistyawati et al, January 2021 0.65 [0.61, 0.69] 1.29
Takoudjou Dzomo et al, January 2021 0.30 [0.28, 0.32] 1.30
Tawalbeh et al, January 2021 0.78 [0.76, 0.80] 1.30
Yoseph et al, January 2021 0.65 [0.62, 0.68] 1.30
Abbasi-Kangevari et al, February 2021 0.86 [0.85, 0.87] 1.31
Ahdab et al, February 2021 0.74 [0.71, 0.77] 1.30
Al-Hussami et al, February 2021 0.79 [0.77, 0.82] 1.30
Lee et al, February 2021 0.87 [0.85, 0.89] 1.30
Masoud et al, February 2021 0.87 [0.86, 0.88] 1.31
Mohamed et al, February 2021 0.86 [0.84, 0.88] 1.30
Naqid et al, February 2021 0.51 [0.48, 0.54] 1.30
Saeed et al, March 2021 0.83 [0.80, 0.85] 1.30
Bates et al, April 2021 0.88 [0.87, 0.89] 1.31
Li et al, April 2021 0.40 [0.36, 0.44] 1.29
Molla and Abegaz, April 2021 0.39 [0.34, 0.43] 1.29
Omar and Amer, April 2021 0.53 [0.50, 0.56] 1.30
Sakr et al, April 2021 0.75 [0.73, 0.77] 1.30
Taddese et al, April 2021 0.53 [0.49, 0.57] 1.29
Heterogeneity: τ² = 0.03, I² = 99.67%, H² = 306.91 0.69 [0.65, 0.73]
Test of θi = θj: Q(68) = 16751.34, p = 0.00
Overall 0.70 [0.66, 0.74]
Heterogeneity: τ² = 0.03, I² = 99.71%, H² = 340.24
Test of θi = θj: Q(73) = 19306.07, p = 0.00
Test of group differences: Qb(1) = 7.14, p = 0.01
0.20
0.40
0.60
0.80
1.00
Random-effects REML model

Figure 3: Analysis by quality of the studies




Study
Attitude score with 95% CI
Weight (%)
Good
Azlan et al, May 2020 0.90 [0.89, 0.91] 1.38
Hezima et al, May 2020 0.89 [0.87, 0.91] 1.38
Alahdal et al, June 2020 0.95 [0.94, 0.96] 1.38
Honarvar et al, June 2020 0.61 [0.58, 0.64] 1.37
Hager et al, July 2020 0.60 [0.57, 0.63] 1.37
Alhazmi et al, August 2020 0.87 [0.85, 0.89] 1.38
Domiati et al, August 2020 0.67 [0.63, 0.72] 1.35
Ghazi et al, September 2020 0.81 [0.77, 0.85] 1.35
Elayeh et al, October 2020 0.59 [0.57, 0.61] 1.38
Yang et al, November 2020 0.93 [0.91, 0.95] 1.38
Banik et al, January 2021 0.76 [0.72, 0.80] 1.36
Shahabi et al, January 2021 0.80 [0.78, 0.82] 1.38
Ahdab et al, February 2021 0.61 [0.57, 0.65] 1.36
Masoud et al, February 2021 0.47 [0.46, 0.48] 1.38
Mohamed et al, February 2021 0.83 [0.81, 0.85] 1.38
Heterogeneity: τ^2 = 0.02, I^2 = 99.65%, H^2 = 283.17 0.75 [0.68, 0.83]
Test of $\theta_i = \theta_j$: Q(14) = 6690.38, p = 0.00
Fair
Zhong et al, March 2020 0.94 [0.93, 0.95] 1.38
Abdelhafiz et al, April 2020 0.76 [0.72, 0.80] 1.36
Al-Hanawi et al, May 2020 0.96 [0.95, 0.97] 1.38
Hayat et al, May 2020 0.75 [0.72, 0.78] 1.37
Alobuia et al, June 2020 0.52 [0.50, 0.54] 1.38
Czeisler et al (USA), June 2020 0.77 [0.75, 0.79] 1.38
La Torre et al, June 2020 0.71 [0.70, 0.72] 1.38
Leehang Lau et al, June 2020 0.67 [0.65, 0.69] 1.38
Sari et al, June 2020 0.87 [0.82, 0.92] 1.35
Afzal et al, July 2020 0.75 [0.72, 0.78] 1.37
Reuben et al, July 2020 0.93 [0.91, 0.96] 1.37
Yue et al, August 2020 0.93 [0.91, 0.95] 1.38
Bates et al, September 2020 0.56 [0.55, 0.57] 1.38
Hossain et al, September 2020 0.51 [0.48, 0.54] 1.37
Khaled et al, September 2020 0.72 [0.69, 0.75] 1.37
Nhu et al, September 2020 0.86 [0.84, 0.88] 1.38
Ngwewondo et al, September 2020 0.66 [0.63, 0.69] 1.37
Paul et al, September 2020 0.82 [0.80, 0.84] 1.38
Islam et al, October 2020 0.88 [0.85, 0.91] 1.37
Wong et al, October 2020 0.67 [0.63, 0.72] 1.35
Bates et al, November 2020 0.63 [0.59, 0.67] 1.36
Mouchtouri et al, November 2020 0.71 [0.69, 0.73] 1.38
Kasemy et al, December 2020 0.77 [0.76, 0.78] 1.38
Nakhostin-Ansari et al, December 2020 0.92 [0.91, 0.94] 1.38
Desalegn et al, January 2021 0.72 [0.69, 0.75] 1.37
Habib et al, January 2021 0.71 [0.68, 0.74] 1.37
Ladiwala et al, January 2021 0.53 [0.51, 0.56] 1.37
Yoseph et al, January 2021 0.83 [0.80, 0.85] 1.37
Al-Hussami et al, February 2021 0.86 [0.84, 0.88] 1.38
Lee et al, February 2021 0.74 [0.71, 0.77] 1.37
Xu et al, March 2021 0.81 [0.80, 0.82] 1.38
Bates et al, April 2021 0.51 [0.49, 0.53] 1.38
Li et al, April 2021 0.64 [0.60, 0.68] 1.36
Molla and Abegaz, April 2021 0.53 [0.48, 0.58] 1.34
Qalati et al, April 2021 0.63 [0.60, 0.66] 1.37
Sakr et al, April 2021 0.71 [0.69, 0.73] 1.38
Heterogeneity: τ^2 = 0.02, I^2 = 99.48%, H^2 = 191.61 0.74 [0.69, 0.78]
Test of $\theta_i = \theta_j$: Q(35) = 7560.46, p = 0.00
Poor
Rahman & Sathi , April 2020 0.42 [0.37, 0.47] 1.35
Ehoche et al, June 2020 0.83 [0.78, 0.88] 1.34
Amalakanti et al, August 2020 0.70 [0.68, 0.72] 1.38
Aqeel et al, August 2020 0.84 [0.81, 0.86] 1.37
Ashiq et al, August 2020 0.70 [0.65, 0.75] 1.34
Rahman et al, August 2020 0.78 [0.76, 0.80] 1.38
Susilkumar & Vengadassalapathy, August 2020 0.91 [0.89, 0.93] 1.38
Adesegun et al, September 2020 0.66 [0.63, 0.69] 1.37
Alabed et al, October 2020 0.74 [0.70, 0.78] 1.36
Ferdous et al, October 2020 0.90 [0.89, 0.91] 1.38
Kuhangana et al, October 2020 0.42 [0.36, 0.47] 1.33
Gao et al, November 2020 0.98 [0.97, 0.99] 1.38
Hossain MJ et al, November 2020 0.50 [0.47, 0.53] 1.37
Okello et al, December 2020 0.51 [0.46, 0.56] 1.34
Sulistyawati et al, January 2021 0.75 [0.72, 0.78] 1.37
Takoudjou Dzomo et al, January 2021 0.61 [0.59, 0.63] 1.38
Tawalbeh et al, January 2021 0.91 [0.90, 0.92] 1.38
Abbasi-Kangevari et al, February 2021 0.93 [0.92, 0.94] 1.38
Naqid et al, February 2021 0.63 [0.60, 0.66] 1.37
Omar and Amer, April 2021 0.80 [0.78, 0.82] 1.37
Heterogeneity: τ^2 = 0.03, I^2 = 99.68%, H^2 = 311.71 0.73 [0.65, 0.80]
Test of $\theta_i = \theta_j$: Q(19) = 3982.04, p = 0.00
Overall 0.74 [0.70, 0.77]
Heterogeneity: τ^2 = 0.02, I^2 = 99.59%, H^2 = 246.28
Test of $\theta_i = \theta_j$: Q(70) = 19302.17, p = 0.00
Test of group differences: $Q_b(2)$ = 0.24, p = 0.89
0.40
0.60
0.80
1.00
Random-effects REML model


Study
Practice score with 95% CI
Weight (%)
Good
Azlan et al, May 2020 0.74 [0.73, 0.75] 1.31
Clements JM, May 2020 0.50 [0.47, 0.53] 1.30
Hezima et al, May 2020 0.61 [0.57, 0.65] 1.29
Alahdal et al, June 2020 0.80 [0.79, 0.82] 1.31
Honarvar et al, June 2020 0.36 [0.34, 0.39] 1.30
Narayana et al, July 2020 0.88 [0.87, 0.89] 1.31
Alhazmi et al, August 2020 0.82 [0.80, 0.84] 1.30
Ghazi et al, September 2020 0.84 [0.79, 0.89] 1.29
Elayeh et al, October 2020 0.66 [0.64, 0.68] 1.30
Yang et al, November 2020 0.84 [0.81, 0.86] 1.30
Banik et al, January 2021 0.60 [0.56, 0.64] 1.29
Shahabi et al, January 2021 0.92 [0.91, 0.93] 1.31
Ahdab et al, February 2021 0.74 [0.71, 0.77] 1.30
Masoud et al, February 2021 0.87 [0.86, 0.88] 1.31
Mohamed et al, February 2021 0.86 [0.84, 0.88] 1.30
Heterogeneity: τ^2 = 0.03, I^2 = 99.72%, H^2 = 363.17 0.74 [0.66, 0.82]
Test of $\theta_i = \theta_j$: Q(14) = 3169.01, p = 0.00
Fair
Zhong et al, March 2020 0.97 [0.96, 0.98] 1.31
Naser et al, April 2020 0.68 [0.66, 0.71] 1.30
Al-Hanawi et al, May 2020 0.87 [0.86, 0.88] 1.31
Hayat et al, May 2020 0.85 [0.83, 0.87] 1.30
Kebede et al, May 2020 0.27 [0.21, 0.33] 1.27
Alobuia et al, June 2020 0.37 [0.35, 0.39] 1.30
Czeisler et al (USA), June 2020 0.55 [0.53, 0.57] 1.30
Leehang Lau et al, June 2020 0.64 [0.61, 0.67] 1.30
Seal et al, June 2020 0.57 [0.54, 0.59] 1.30
Afzal et al, July 2020 0.65 [0.62, 0.68] 1.30
Reuben et al, July 2020 0.74 [0.70, 0.78] 1.29
Yue et al, August 2020 0.68 [0.64, 0.72] 1.29
Bates et al, September 2020 0.93 [0.92, 0.95] 1.31
Hossain et al, September 2020 0.80 [0.79, 0.82] 1.31
Khaled et al, September 2020 0.78 [0.75, 0.81] 1.30
Nhu et al, September 2020 0.93 [0.92, 0.94] 1.31
Ngwewondo et al, September 2020 0.81 [0.79, 0.83] 1.30
Paul et al, September 2020 0.77 [0.75, 0.79] 1.30
Sengeh et al, September 2020 0.69 [0.66, 0.71] 1.30
Islam et al, October 2020 0.82 [0.78, 0.85] 1.29
Leigh et al, October 2020 0.57 [0.55, 0.59] 1.30
Wong et al, October 2020 0.67 [0.63, 0.72] 1.29
Bates et al, November 2020 0.92 [0.90, 0.95] 1.30
Mouchtouri et al, November 2020 0.57 [0.55, 0.59] 1.30
Kasemy et al, December 2020 0.53 [0.52, 0.55] 1.31
Nakhostin-Ansari et al, December 2020 0.94 [0.92, 0.95] 1.31
Bekele et al, January 2021 0.52 [0.47, 0.57] 1.27
Desalegn et al, January 2021 0.62 [0.58, 0.66] 1.29
Habib et al, January 2021 0.65 [0.62, 0.68] 1.30
Ladiwala et al, January 2021 0.72 [0.69, 0.74] 1.30
Yoseph et al, January 2021 0.65 [0.62, 0.68] 1.30
Al-Hussami et al, February 2021 0.79 [0.77, 0.82] 1.30
Lee et al, February 2021 0.87 [0.85, 0.89] 1.30
Saeed et al, March 2021 0.83 [0.80, 0.85] 1.30
Xu et al, March 2021 0.78 [0.77, 0.79] 1.31
Bates et al, April 2021 0.88 [0.87, 0.89] 1.31
Li et al, April 2021 0.40 [0.36, 0.44] 1.29
Molla and Abegaz, April 2021 0.39 [0.34, 0.43] 1.29
Qalati et al, April 2021 0.63 [0.60, 0.66] 1.30
Sakr et al, April 2021 0.75 [0.73, 0.77] 1.30
Heterogeneity: τ^2 = 0.03, I^2 = 99.67%, H^2 = 299.11 0.70 [0.65, 0.75]
Test of $\theta_i = \theta_j$: Q(39) = 9913.78, p = 0.00
Poor
Rahman & Sathi , April 2020 0.93 [0.91, 0.96] 1.30
Amalakanti et al, August 2020 0.77 [0.75, 0.79] 1.30
Aqeel et al, August 2020 0.93 [0.92, 0.95] 1.31
Ashiq et al, August 2020 0.57 [0.51, 0.63] 1.27
Haq et al, August 2020 0.59 [0.54, 0.64] 1.28
Rahman et al, August 2020 0.74 [0.71, 0.77] 1.30
Susilkumar & Vengadassalapathy, August 2020 0.88 [0.86, 0.90] 1.30
Adesegun et al, September 2020 0.60 [0.57, 0.63] 1.30
Alabed et al, October 2020 0.34 [0.30, 0.38] 1.29
Ferdous et al, October 2020 0.76 [0.74, 0.78] 1.30
Gao et al, November 2020 0.97 [0.96, 0.98] 1.31
Hossain MJ et al, November 2020 0.62 [0.60, 0.64] 1.30
Okello et al, December 2020 0.48 [0.43, 0.53] 1.28
Sulistyawati et al, January 2021 0.65 [0.61, 0.69] 1.29
Takoudjou Dzomo et al, January 2021 0.30 [0.28, 0.32] 1.30
Tawalbeh et al, January 2021 0.78 [0.76, 0.80] 1.30
Abbasi-Kangevari et al, February 2021 0.86 [0.85, 0.87] 1.31
Naqid et al, February 2021 0.51 [0.48, 0.54] 1.30
Omar and Amer, April 2021 0.53 [0.50, 0.56] 1.30
Taddese et al, April 2021 0.53 [0.49, 0.57] 1.29
Heterogeneity: τ^2 = 0.04, I^2 = 99.70%, H^2 = 330.67 0.67 [0.58, 0.75]
Test of $\theta_i = \theta_j$: Q(19) = 6335.58, p = 0.00
Overall 0.70 [0.66, 0.74]
Heterogeneity: τ^2 = 0.03, I^2 = 99.70%, H^2 = 334.37
Test of $\theta_i = \theta_j$: Q(74) = 19477.71, p = 0.00
Test of group differences: $Q_b(2)$ = 1.32, p = 0.52
0.20
0.40
0.60
0.80
1.00
Random-effects REML model